

UNIT – 2

DC MOTORS

Principle of Operation

DC motor operates on the principle that when a current carrying is placed in a magnetic field, it experiences a mechanical force given by $F = BIL$ newton. Where 'B' = flux density in wb/m^2 , 'I' is the current and 'L' is the length of the conductor. The direction of force can be found by Fleming's left hand rule. Constructionally, there is no difference between a DC generator and DC motor.

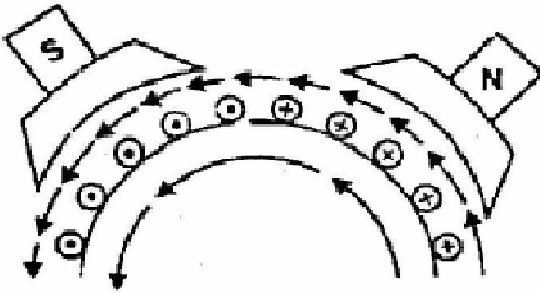


Figure 2.1

Figure 2.1 shows a multipolar DC motor. Armature conductors are carrying current downwards under North Pole and upwards under South Pole. When the field coils are excited, with current carrying armature conductors, a force is experienced by each armature conductor whose direction can be found by Fleming's left hand rule. This is shown by arrows on top of the

conductors. The collective force produces a driving torque which sets the armature into rotation.

The function of a commutator in DC motor is to provide a continuous and unidirectional torque.

In DC generator the work done in overcoming the magnetic drag is converted into electrical energy. Conversion of energy from electrical form to mechanical form by a DC motor takes place by the work done in overcoming the opposition which is called the 'back emf'.

BACK EMF: is the dynamically induced emf in the armature conductors of a dc motor when the armature is rotated. The direction of the induced emf as found by Flemings right hand rule is in opposition to the applied voltage. Its value is same as that of the induced emf in a DC generator i.e., $E_b = \left(\frac{\phi Z n}{60}\right) \times \frac{P}{A}$ volts. This emf is called as back emf E'_b . The work done in overcoming this opposition is converted into mechanical energy.

SIGNIFICANCE OF BACK EMF: Figure 2.2 shows a DC shunt motor. The rotating armature generating the back emf E'_b is like a battery of emf E'_b connected across a supply voltage of 'V' volts.

From Figure 2.2, $I_a = \frac{V - E_b}{r_a}$ where r_a = armature resistance.

$$E_b = \frac{\phi Z N P}{60 A} \text{ Volts. } E_b \propto N.$$

If E_b is large, armature current will be less and vice versa. Hence E_b acts like a governor i.e., it makes the motor self-regulating so that it draws as much current as required by the motor.

VOLTAGE EQUATION OF A MOTOR:

The voltage 'V' applied across the motor armature has to

- i) Overcome the back emf E_b and
- ii) Supply the armature ohmic drop $I_a r_a$

Hence $V = E_b + I_a r_a \dots 1$ This is known as voltage equation of DC motor.

Multiplying both sides of voltage equation by I_a

$$VI_a = E_b I_a + I_a^2 r_a \dots 2$$

$VI_a = \text{electrical input to the armature.}$

$E_b I_a = P_m = \text{electrical equivalent of mechanical power developed in the armature.}$

$I_a^2 r_a = \text{armature copper loss}$

CONDITION FOR MAXIMUM POWER

Mechanical power developed by the motor is $P_m = VI_a - I_a^2 r_a \dots 3$

Condition for maximum power is $\frac{dP_m}{dI_a} = 0$

$$\frac{dP_m}{dI_a} = V - 2I_a r_a = 0$$

$$\frac{V}{2} = I_a r_a$$

$$V = E_b + \frac{V}{2} \quad E_b = \frac{V}{2} \dots 4$$

Thus the mechanical power developed by the motor is maximum when the back emf is equal to half the applied voltage.

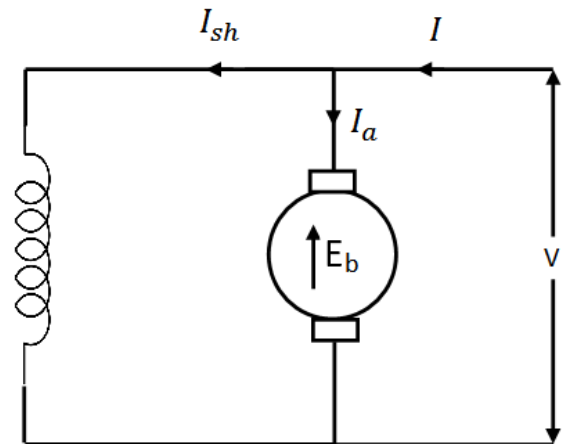
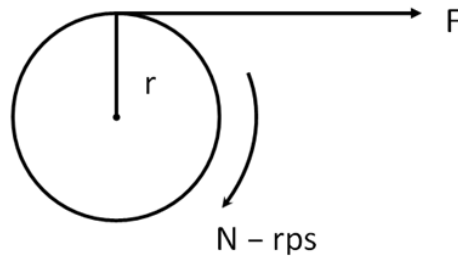


Figure 2.2

TORQUE:

Torque is the twisting moment about an axis. It is measured by the product of the force and the radius at which the force acts. Consider a pulley of radius 'r' metre acted upon a circumferential force of 'F' newton which causes it to rotate at 'N' rotations per second (r.p.s) as show in Figure. 2.3

**Figure 2.3**

$$T = F \times r \text{ N-metre}$$

Work done by this force in one revolution = Force X Distance = $F \times 2\pi r$ joule.

Power Developed = $F \times 2\pi r \times N$ joule/sec.

$$= (F \times r) 2\pi N \text{ joule/sec}$$

$2\pi N$ = angular revolution ω in rad/sec. And $F \times r$ = Torque 'T'

Therefore Power Developed = $T \times \omega$ joule/sec or watts..... 5

ARMATURE TORQUE OF A MOTOR:

Let T_a = armature torque in N –m developed by the armature of a motor running at N.rps.

Therefore $P = T_a \times 2\pi N$ Watts 6

Electrical equivalent of mechanical power developed $P_m = E_b I_a$

$$E_b I_a = T_a \times 2\pi N$$

$$T_a = \frac{1}{2\pi} \frac{E_b I_a}{N} \text{ N – m7}$$

$$= 0.0162 \frac{E_b I_a}{N} \text{ kg-m8}$$

Also, on substituting for E_b i.e., $\Phi ZN \left(\frac{P}{A}\right) I_a$

$$\Phi ZN \left(\frac{P}{A}\right) I_a = T_a \times 2\pi N$$

Therefore $T_a = \frac{1}{2\pi} \Phi Z I_a \left(\frac{P}{A}\right) \text{ N-mtrs}$

$$T_a = 0.159 \Phi Z I_a \left(\frac{P}{A}\right) \text{ N-mtrs} \dots\dots 9$$

$$= 0.162 \Phi Z I_a \left(\frac{P}{A}\right) \text{ kg-mtrs} \dots\dots\dots 10$$

$$(1 \text{ Kg} = 9.81 \text{ N})$$

From the above equation for torque, it is seen that

- (i) $T \propto \Phi I_a$
- (ii) $T \propto I_a^2$ - For series motor (because $\Phi \propto I_a$) before saturation. After saturation $T \propto I_a$.
- (iii) $T \propto I_a$ - For shunt motor. (because Φ is constant in a shunt motor)

SHAFT TORQUE (T_{sh}):

Some of the torque developed in the armature will be lost in supplying the iron and friction losses in the motor. The torque which is available for doing useful work is known as shaft torque ' T_{sh} '. The horse power obtained by using the shaft torque is called as Brake Horse Power (BHP).

$$\text{BHP} = \frac{T_{sh} \times 2\pi N}{735.5} \dots\dots\dots 11$$

$$T_{sh} = \frac{\text{BHP} \times 735.5}{2\pi N} = \frac{\text{output in watts}}{2\pi N} \text{ N-m}$$

$$\text{Lost torque} = T_a - T_{sh}$$

$$= 0.159 \times \frac{\text{Iron and Friction Losses in Watts}}{N} \text{ N-mtrs} \dots\dots\dots 12$$

$$= 0.162 \times \frac{\text{Iron and Friction Losses in Watts}}{N} \text{ kg-mtrs} \dots\dots\dots 13$$

SPEED OF A DC MOTOR:

$$E_b = V - I_a r_a \quad \text{Or} \quad \Phi Z N \left(\frac{P}{A}\right) = V - I_a r_a$$

$$N = \frac{E_b}{\Phi Z} \left(\frac{A}{P}\right) \text{ r.p.s}$$

$$N = K \frac{E_b}{\Phi}$$

$$N \propto \frac{E_b}{\Phi}$$

$$T_a \propto \Phi I_a$$

$$\Phi = \text{Constant, if } \Phi = \text{constant, then } T_a \propto I_a \dots\dots\dots 14$$

(i) For series motor, $N_1 \propto \frac{E_{b1}}{\Phi_1}$

$$N_2 \propto \frac{E_{b2}}{\Phi_2} \text{ where } E_{b1} = V - I_{a1}r_a$$

$$E_{b2} = V - I_{a2}r_a$$

$$\frac{N_2}{N_1} = \frac{E_{b2} I_{a1}}{E_{b1} I_{a2}} \text{ Or } \frac{N_2}{N_1} = \frac{E_{b2} \Phi_1}{E_{b1} \Phi_2} \dots\dots\dots 15$$

(ii) For shunt motor.

$$\frac{N_2}{N_1} = \frac{E_{b2} \Phi_1}{E_{b1} \Phi_2}$$

$\Phi = \text{constant}$ for shunt motor

$$\Phi_1 = \Phi_2$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \dots\dots\dots 16$$

Speed regulation: is defined as the change in speed from No-load to full load when the rated load on the motor is reduced to zero, expressed as a percentage of rated speed.

$$\% \text{ speed regulation} = \frac{\text{NoLoadspeed} - \text{Full loadspeed}}{\text{FullLoadspeed}} = \frac{dN}{N} \times 100$$

CHARACTERISTICS OF DC MOTORS:

There are three important characteristics.

1. Armature torque vs Armature current T_a vs I_a (Electrical characteristics)
2. Speed vs armature current characteristic
3. Speed vs torque N vs T_a (Mechanical characteristics)

CHARACTERISTICS OF SHUNT MOTORS

i) $\frac{T_a}{I_a}$

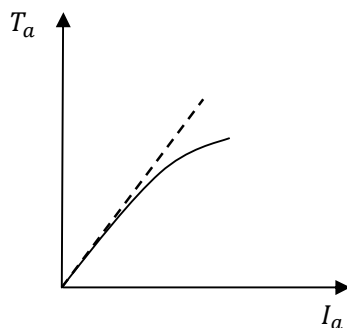


Figure 2.4

ii) $\frac{N}{I_a}$

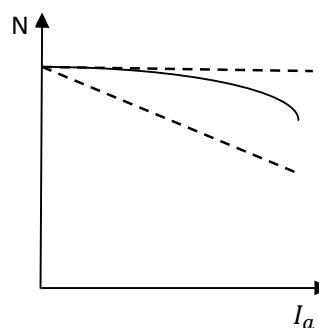


Figure 2.5

iii) $\frac{N}{T_a}$

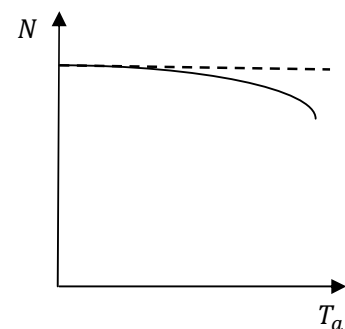


Figure 2.6

$$N \propto \frac{E_b}{\phi} \text{ OR } N = K_N \left(\frac{V - I_a R_a}{\phi} \right) ; N = \frac{K_N V}{\phi} - \frac{K_N I_a r_a}{\phi} \text{----- (a)}$$

$$I_a = \frac{T}{K_T \phi} \text{----- (b)}$$

Substituting (b) in (a)

$$N = \frac{K_N V}{\phi} - \frac{K_N r_a}{K_T \phi^2} T \text{----- (17)}$$

1. T_a vs I_a CHARACTERISTICS

For a shunt motor flux ϕ can be assumed practically constant (at heavy loads, ϕ decreases, due to increased armature reaction)

$$T_a \propto \phi I_a$$

$$\phi = \text{constant}, T_a \propto I_a$$

Therefore electrical characteristic of a shunt motor is a straight line through origin shown by dotted line in Figure 2.4. Armature reaction weakens the flux hence T_a vs I_a characteristic bends as shown by dark line in figure 2.4, Shunt motors should never be started on heavy loads, since it draws heavy current under such condition.

2. N vs I_a CHARACTERISTICS

$$N \propto E_b ; \quad E_b \text{ is practically almost constant.}$$

Hence the speed is constant. However, E_b decreases slightly more than ϕ with increase in load and thus there is slight decrease in speed. This decrease in speed varies from 5 to 15% of full load speed and it depends on armature reaction and saturation. This characteristics is shown in Figure 2.5

3. N vs T_a CHARACTERISTICS

This characteristic can be deduced from 1 and 2, shown in figure 2.6

Performance curves of DC shunt motor

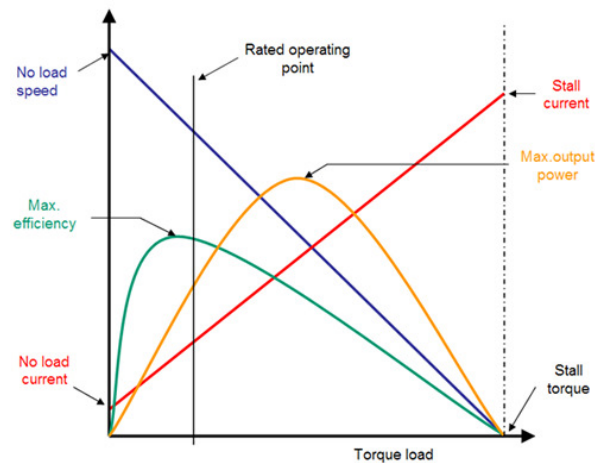


Figure 2.7

The four essential characteristics of shunt motor i.e., torque, speed, current and efficiency plotted as a function of horse power are known as performance curves of the motor, shown in figure 2.7

Shunt motor has a definite No-load speed hence can be used where a load is suddenly thrown off with field circuit remaining close. The drop in speed from No-load to full load is small and hence referred to as constant speed motor. The efficiency curve is usually of the same shape for all motors and generators. It is advantageous to have an efficiency curve which is fairly flat and the maximum efficiency near to full load. Certain value of minimum current is required even when the output is zero as the input under No-load condition has to meet the losses within the machine.

The shunt motor is also capable of starting under heavy load condition but the current drawn by the motor will be very high ($T_a \propto I_a$) compared to DC series motor. As the series motor draws only one and half times the full load current $T_a \propto I_a^2$, $I_a \propto \sqrt{T_a}$.

CHARACTERISTICS OF SERIES MOTORS

1. T_a vs I_a CHARACTERISTICS

$$T_a \propto \phi I_a$$

$$\phi \propto I_a$$

$$T_a \propto I_a^2 - \text{upto saturation} \dots\dots\dots 18$$

$$T_a \propto I_a - \text{after saturation} \dots\dots\dots 19$$

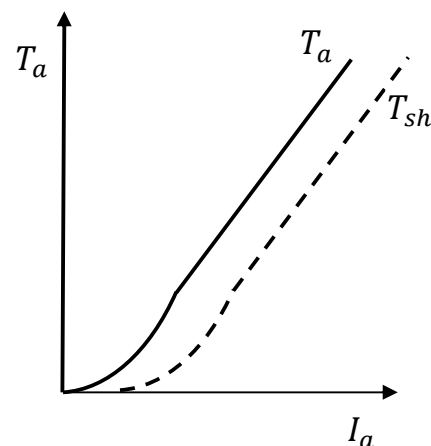


Figure 2.8

At light loads, I_a and hence ϕ is small. But as I_a increases T_a increases as the square of the current up to saturation. After saturation ϕ becomes constant, the characteristic becomes a straight line as shown in Figure 2.8. Therefore a series motor develops a torque proportional to the square of the armature current. This characteristic is suited where huge starting torque is required for accelerating heavy masses.

Ex. Hoists, electric trains, etc.

2. N vs I_a CHARACTERISTICS

$$N \propto \frac{E_b}{\phi} \quad E_b \text{ is approximately constant}$$

$$N \propto \frac{1}{\phi} \dots\dots\dots 20$$

If I_a increases, ϕ increases and hence speed decreases.

This characteristic is shown in figure 2.9(a). Change in E_b for various load currents is small. Hence may be neglected. Therefore the speed is inversely proportional to flux, because $N \propto \frac{E_b}{\phi}$. When the load is heavy, I_a is large and speed is low. When the load is low, current and hence flux will be small. Therefore speed becomes dangerously high. Hence a series motor should never be started without load on it.

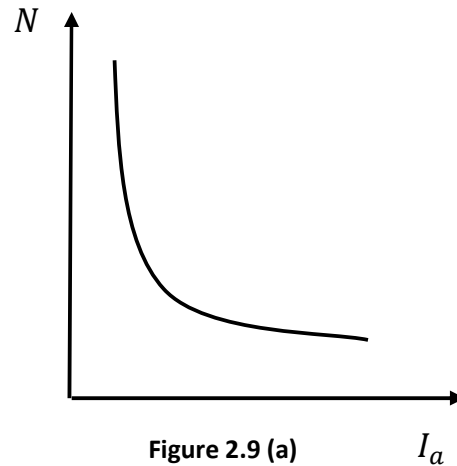


Figure 2.9 (a)

3. N vs T_a CHARACTERISTICS

$$T_a \propto E_b I_a$$

$$E_b \text{ is constant}$$

Hence, $T_a \propto I_a$. Therefore, N vs T_a characteristic

can be deduced from 1 and 2 as shown in Figure 2.9 (b)

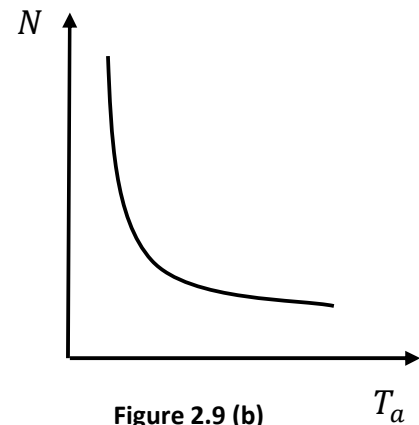


Figure 2.9 (b)

PERFORMANCE CURVES OF DC SERIES MOTOR

The performance curves of DC series motor are shown in Figure 2.10. The machine is so designed that it is having maximum efficiency near rated load.

For a given input current, the starting torque developed by a DC series motor is greater than that developed by a shunt motor. Hence series motors are used where huge starting torques are necessary. Ex. Cranes, hoists, electric traction etc. The DC series motor responds by decreasing its speed for the increased in load. The current drawn by the DC series motor for the given increase in load is lesser than DC shunt motor. The drop in speed with increased load is much more prominent in series motor than that in a shunt motor. Hence series motor is not suitable for applications requiring a constant speed.

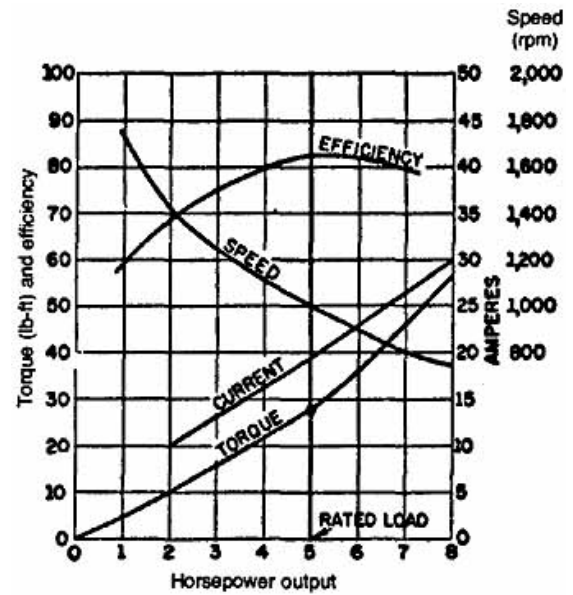


Figure 2.10

COMPOUND MOTOR CHARACTERISTICS

Cumulative compound motors: are used where series characteristics are required and in addition the load is likely to be removed totally such as in some types of coal cutting machines or for driving heavy machine tools which have to take sudden deep cuts quite often. Speed will not become excessively high due to shunt winding and the motor will be able to take heavy loads because of series winding.

Differential compound motors: Series field opposes the shunt field; therefore the flux is decreased as the load is applied to the motor. This results in the motor speed remaining almost constant or even increasing with increase in load.

Summarizing,

1. $\frac{N}{I_a}$ Lies between shunt ($\phi = \text{constant}$) and series ($\phi \propto I_a$) characteristics as shown in figure 2.11.
2. Used in rolling mills where light and heavy loads are thrown on the motor.

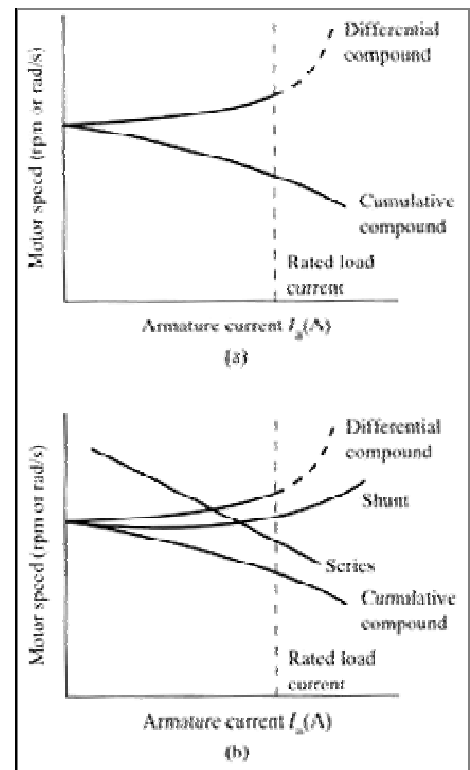


Figure 2.11

3. Cumulative compound motor has a finite No-load speed and has a load – relieving characteristics of a series motor under heavy load conditions.
4. Differential compound motors are seldom used in practice.(because of rising speed characteristics)

SPEED CONTROL OF DC MOTORS

DC motors are in general much more adaptable speed drives than AC motors. Speed of a DC motor can be controlled in a wide range.

$$N = K_N \frac{E_b}{\phi} = K_N \frac{(V - I_a r_a)}{\phi} \text{ Since armature resistance drop is small, } N = K_N \frac{V}{\phi} \dots\dots\dots 21$$

From the above equation, two methods of speed control are possible.

1. Variation of field current which varies the flux/pole (Φ) and is known as field control.
2. Variation of terminal voltage 'V' known as armature voltage control.

FIELD CONTROL:

$$\text{For a fixed terminal voltage, } \frac{N_2}{N_1} = \frac{\Phi_1}{\Phi_2} = \frac{I_{f1}}{I_{f2}} \text{ (for linear magnetization) } \dots\dots\dots 22$$

Limitations of speed control by field control:

1. 'N' below rated speed is not possible. Because Φ can be decreased and cannot increase.
2. $N \propto \frac{1}{\phi}$ & $T \propto \phi$ for a given armature current, this method suits for constant kW drives only where 'T' decreases if speed decreases.
3. Not suited for speed reversal.

DC SHUNT MOTOR:

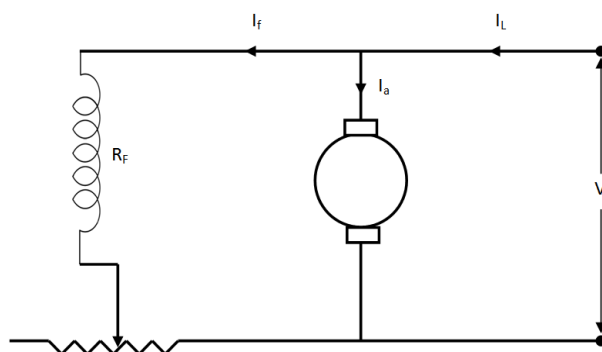


Figure 2.12

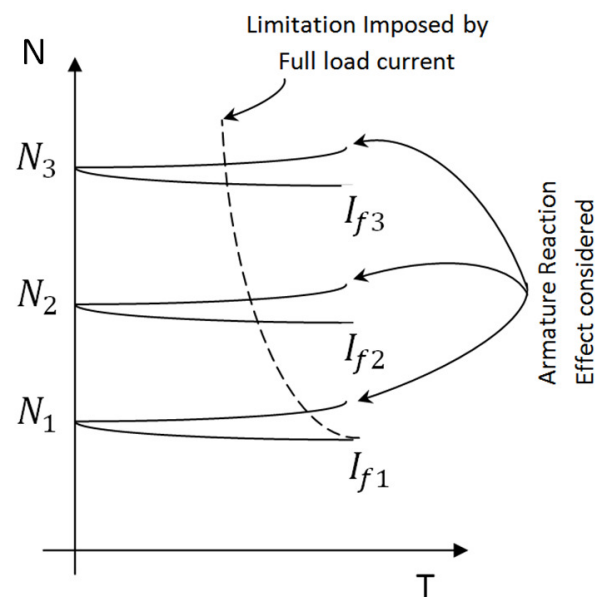


Figure 2.13

$$N = \frac{K_N V}{\phi} - \left(\frac{K_N r_a}{K_T \phi^2} \right) T \dots\dots\dots 23$$

Speed control is achieved by means of a rheostat in the field circuit as shown in the Figure 2.12. The speed torque characteristics have a small linear drop due to the second term (r_a effect) in the above equation. The field is weakened due to the armature reaction. The working range of the speed torque characteristics reduces with increasing speed in order for the armature current not to exceed the full load value with a weakening field. The speed – torque curves are shown in Figure 2.13

DC SERIES MOTOR:

Speed control is achieved by adjusting the field ampere turns. There are three ways for varying the field ampere turns.

1. DIVERTER FIELD CONTROL:

Diverter resistance R_d is connected across the field winding as shown in Figure 2.14. By varying R_d the field current and hence the field ampere turns can be reduced. Therefore, the field current is given by

$$I_f = I_a \left(\frac{R_d}{R_d + R_{se}} \right) K_d I_a \dots \dots 24$$

$$\text{Where } K_d = \frac{R_d}{R_d + R_{se}}$$

$$K_d = \frac{1}{1 + \frac{R_{se}}{R_d}}$$

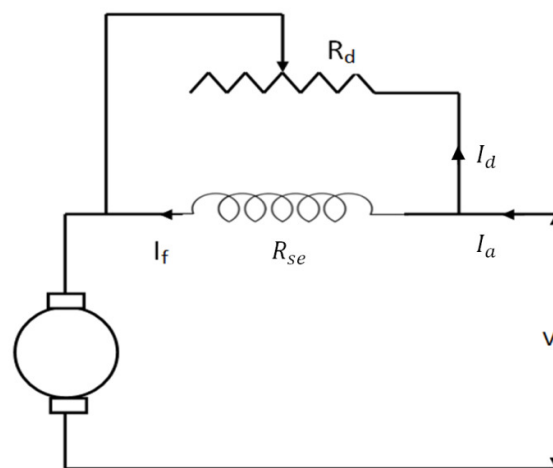


Figure 2.14

We know that, for series motor $N = \frac{V - I_a(r_a + r_{se})}{\Phi}$

And $T \propto \Phi I_a$ or $T = K_T \Phi I_a$

$$T = K_T K_f I_a^2, \quad I_a = \sqrt{\frac{T}{K_T K_f}}$$

If there is no R_d , then $\Phi = K_f I_a$

$$N = K_N \left[\frac{V}{\Phi} - \frac{I_a(r_a + r_i)}{\Phi} \right]$$

$$N = K_N \left[\frac{V}{K_f I_a \sqrt{K_T} \sqrt{K_f} \cdot K_f \frac{\sqrt{T}}{\sqrt{K_T} \sqrt{K_f}}} - \frac{\sqrt{T}(r_a + r_{se})}{K_f I_a \sqrt{K_T} \sqrt{K_f} \cdot K_f \frac{\sqrt{T}}{\sqrt{K_T} \sqrt{K_f}}} \right]$$

$$N = \frac{K_N}{K_f} \left[\frac{V}{I_a} - (r_a + r_{se}) \right] \dots\dots\dots 25$$

If R_d is introduced then,

$$\Phi = K_d K_f I_a.$$

$$\text{Therefore } N = K_N \left[\frac{V}{\Phi} - \frac{I_a(r_a + r_{se})}{\Phi} \right]$$

$$K_N \left[\frac{V}{K_d K_f I_a} - \frac{I_a(r_a + r_{se} || R_d)}{K_d K_f I_a} \right]$$

$$\text{Substituting } I_a = \sqrt{\frac{T}{K_d K_f}}$$

$$N = \frac{K_N}{K_f} \left[\frac{\sqrt{K_f K T}}{\sqrt{T} \sqrt{K_d}} - \frac{(r_a + r_{se} || R_d)}{K_d} \right] \dots\dots\dots 26$$

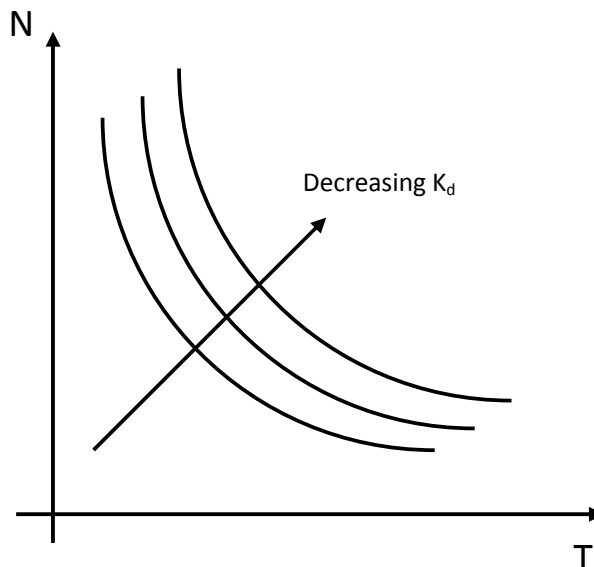


Figure 2.15

From the above equation speed torque characteristics for decreasing values of K_d are plotted which is shown in above Figure 2.15.

2. Tapped field control:

The field ampere turns are adjusted in steps by varying the number of turns included in the circuit as shown in figure 2.16, the following relations are obtained from the circuit.

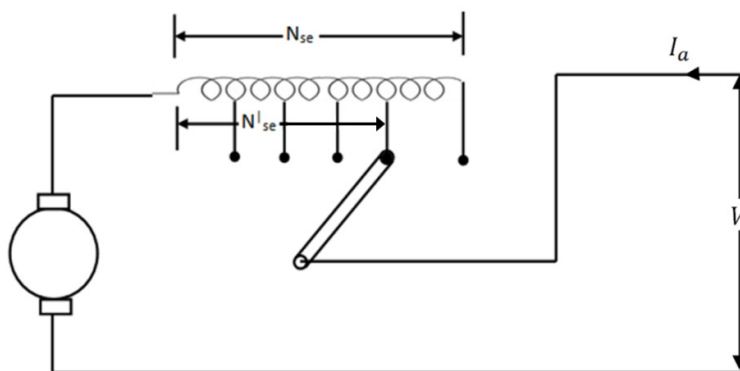


Figure 2.16

$$\text{AT (effective)} = \frac{N_{se}}{N_{se}} I_a = K_{se} I_a$$

$$\text{Therefore } \Phi = K_f K_{se} I_a \dots\dots\dots 27$$

$$N_{se}^I = \text{turns with } r_{se}^I = K_{se} r_{se}$$

$$T = K_f K_{se} K_t I_a^2$$

$$I_a = \sqrt{\frac{T}{K_T K_f K_{se}}} \dots\dots\dots 28$$

$$\text{From equation 25, } N = \frac{K_N}{K_f} \left[\frac{V}{I_a} - (r_a + r_{se}) \right]$$

$$\text{Substituting } I_a = \sqrt{\frac{T}{K_T K_f K_{se}}},$$

$$N = \frac{K_N}{K_f} \left[\frac{V \sqrt{K_T K_f K_{se}}}{\sqrt{T}} - (r_a + K_{se} r_{se}) \right]$$

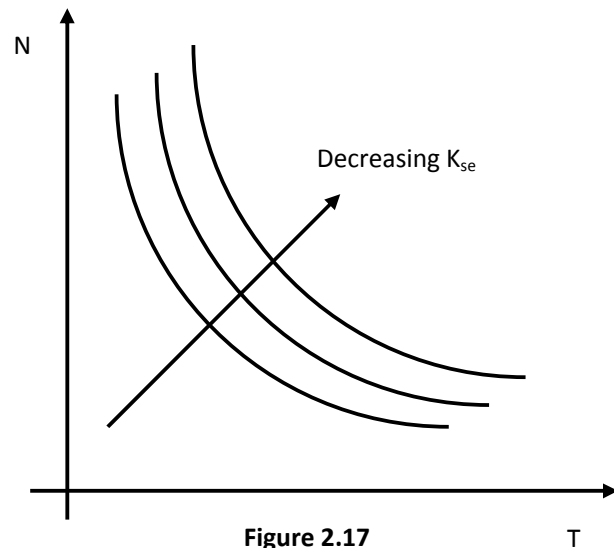


Figure 2.17

The speed torque characteristics are shown in figure 2.17 and are of similar to those of diverter field control.

3. Series parallel control:

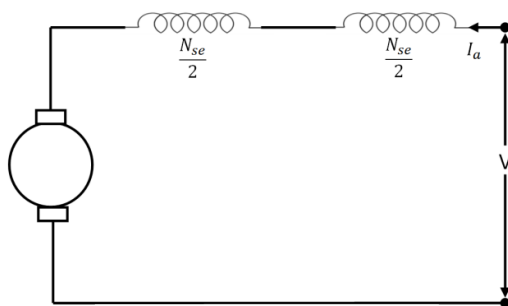


Figure 2.18

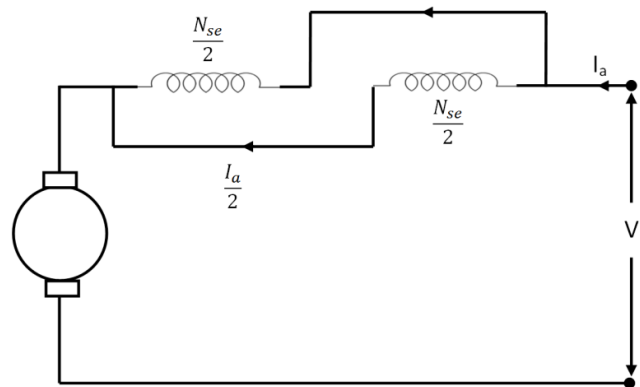


Figure 2.19

In this method, the field windings are divided into two equal halves and then connected in series or parallel to control the field ampere turns as shown in figure 2.18 and 2.19 respectively.

$$2 \left(\frac{N_{se}}{2} \times \frac{I_a}{2} \right) = AT \text{ parallel} \dots\dots\dots 30$$

$$AT \text{ parallel} = \frac{1}{2} (AT_{series}) \dots\dots\dots 31$$

$$\Phi = \frac{1}{2} K_f I_a$$

$K_{se} = 1$ or $\frac{1}{2}$ i.e., only two speeds are possible; parallel field connection gives the higher speed, series field connection gives the lower speed.

COMPARISON OF TORQUE DEVELOPED BY MOTORS IN SERIES AND PARALLEL CONNECTION:

MOTORS IN SERIES:

$$N \propto \frac{V}{I} \propto \frac{V}{2I} \dots\dots\dots 32$$

This speed is $1/4^{\text{th}}$ of the speed of the motor, when in parallel.

$$T \propto \Phi I \propto I^2 \dots\dots\dots 33$$

Torque is four times that produced by motors when in parallel.

MOTORS IN PARALLEL:

$$N \propto \frac{E_b}{\Phi} \propto \frac{E_b}{I_a} \text{ (series motor)}$$

Since $E_b \cong V$; $N \propto \frac{V}{I} \propto \frac{2V}{I}$; Torque $\propto \Phi I \propto I^2$;

$$\text{Therefore } T \propto \left(\frac{I}{2}\right)^2; T \propto \frac{I^2}{4} \dots\dots\dots 34$$

From equations 33 and 34 torque developed in series connection is four times that of motors in parallel connection.

ARMATURE VOLTAGE CONTROL:

In this method, applied voltage across the armature of the DC motor is varied. This method is superior to field control in the following aspects:

1. This method provides a constant torque drive.(if the Φ and I_a are maximum, maximum torque can be obtained as $T \propto \Phi I_a$)
2. Since main field ampere turns are maintained at large value, flux density distortion caused by armature reaction is limited.
3. Unlike field control scheme, speed reversal can be easily implemented.

This method requires a variable voltage supply which makes this method costlier.

DC SHUNT MOTOR:

Following are the armature control schemes for DC shunt motor.

1. Rheostatic control:

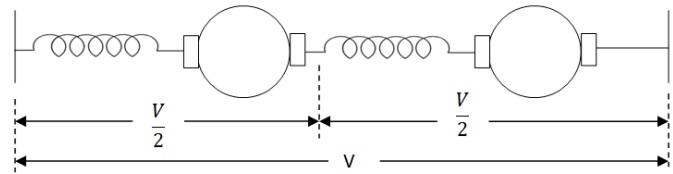


Figure 2.20

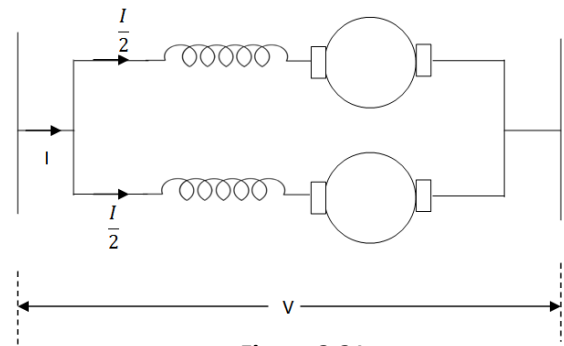


Figure 2.21

Here the applied armature voltage is varied by placing an adjustable resistance 'R' in series with the armature as shown in the Figure 2.22. N vs T for varying 'R' is shown in figure 2.23.

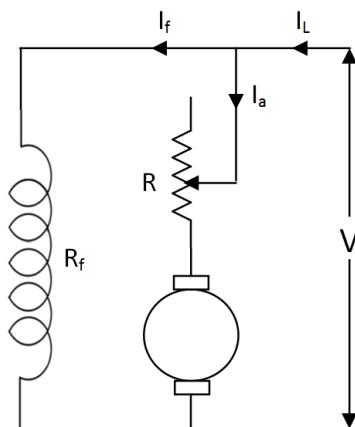


Figure 2.22

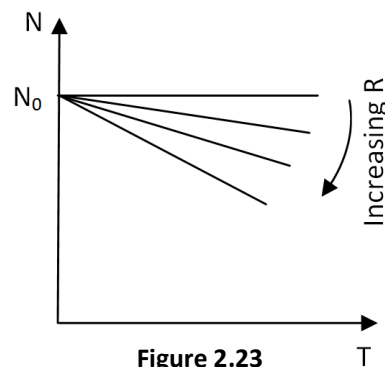


Figure 2.23

Some of the limitations of the rheostatic method are:

1. Speeds only below rated speed
2. Range of speeds is limited because efficiency reduces drastically for large speed reductions

$$\eta_{arm} = \frac{E_b I_a}{V I_a} = \frac{(V - I_a r_e)}{V I_a} = \frac{1 - I_a r_e}{V}$$

3. Speed regulation is poor. Because for a given resistance r_e , N varies directly with load.

Therefore this method is suitable for very small (fractional kW) or for short-time, intermittent slowdowns for medium sized motors.

$$N_0 \propto V - I_{a0} r_a$$

$$N_1 \propto V - I_a (r_a + R)$$

$$\frac{N}{N_0} = \frac{V - I_a (r_a + R)}{V - I_{a0} r_a} \quad \frac{N}{N_0} = \frac{V - I_a (r_a + R)}{V}$$

The load current for which $N=0$ is

$$0 = N_0 \left(1 - \frac{I_{a1} R_t}{V} \right)$$

$$0 = 1 - \frac{I_{a1} R_t}{V}; \quad \frac{I_{a1} R_t}{V} = 1; \quad I_{a1} = \frac{V}{R_t} \dots\dots\dots 35$$

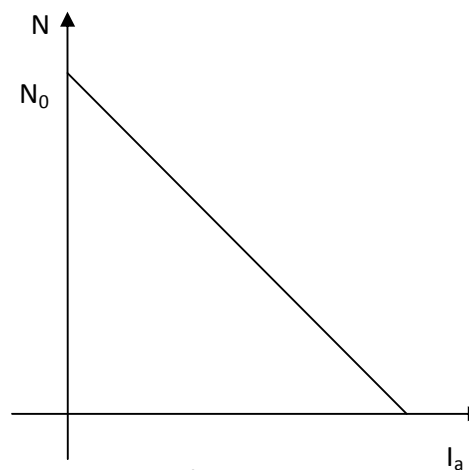


Figure 2.24

This is the maximum current and N_0 is the No-load speed shown in figure 2.24

1. Shunted armature control:

In the armature rheostatic control method, the change in armature current due to change in load will affect the speed. Hence in this method the armature is shunted by an adjustable resistance as shown in Figure 2.25.

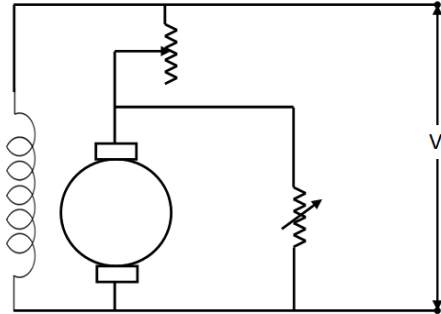


Figure 2.25

Advantages of this method are

- a. Speed regulation will be better.
- b. The changes in the armature current due to load will not be as effective as the armature is connected across a resistance.

SERIES PARALLEL CONTROL OF SHUNT MOTOR:

In this method the two identical motors are coupled together mechanically to a common load. Two speeds at constant torque are possible in this method – one by connecting motor armature in series and the other by connecting them in parallel as shown in figures 2.26 and 2.27 respectively. When connected in series the voltage across each motor is half the supply voltage and when in parallel, each armature carries full supply voltage.

This method is superior to rheostatic control so far as efficiency is concerned but the speeds that can be obtained can only be two steps. This method is commonly employed for speed control of series traction motors.

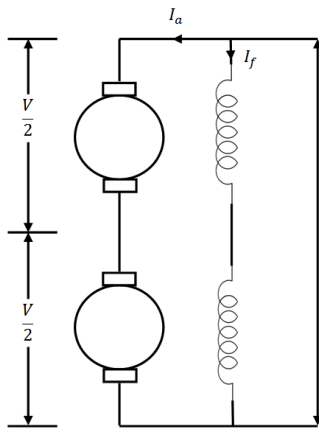


Figure 2.26

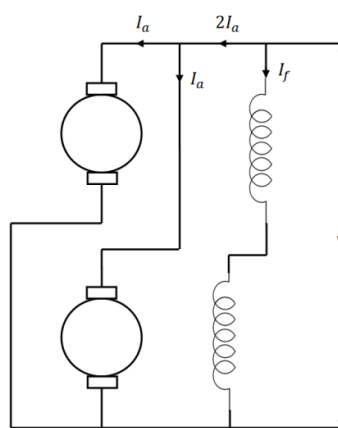
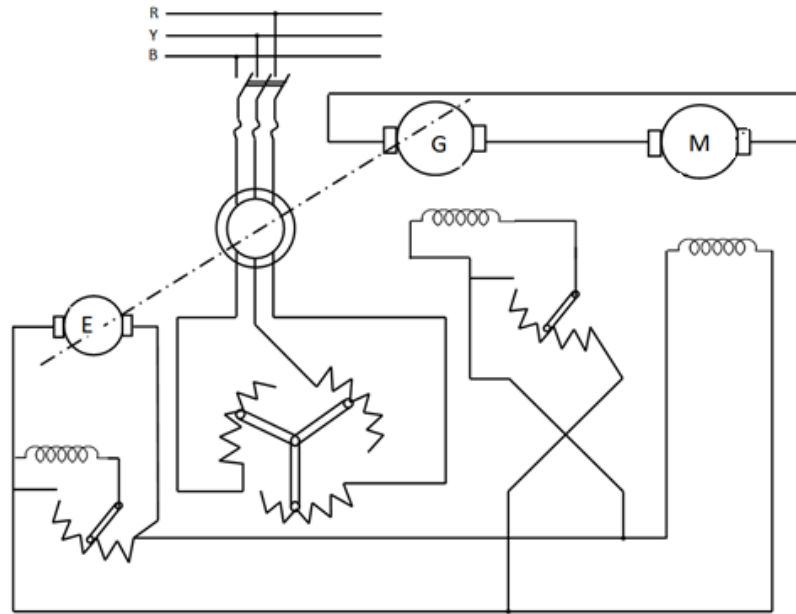
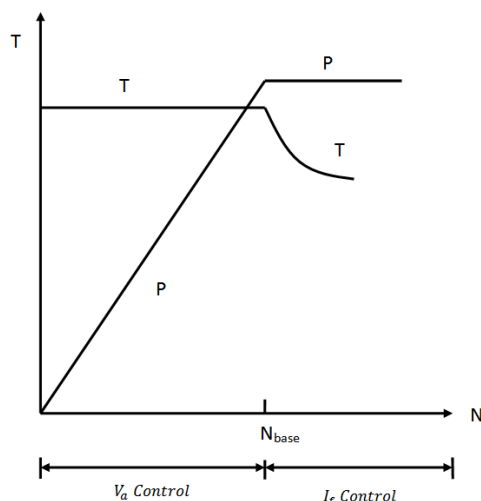


Figure 2.27

WARD-LEONARD METHOD OF SPEED CONTROL:**Figure 2.28**

It is a combined armature and field control and is therefore operationally the most efficient method of speed control with a wide range. 'M' is the main motor whose speed control is required. The field of this motor is permanently connected across the DC supply lines. Its armature is supplied by a variable voltage derived by a polyphase Induction Motor-DC generator set. The field of the DC generator is excited from an exciter coupled to the extension of the induction motor shaft. The entire arrangement is shown in the figure 2.28. The reversible switch provided for the generator makes it possible to easily reverse the generator excitation thereby reversing the voltage polarity for reversing the direction of rotation of motor. Though expensive, this arrangement can be easily adapted to feedback schemes for automatic control of speed. This method provides both constant torque and constant HP (or kW) drive.

**Figure 2.29**

The motor armature and its field winding are fed at maximum values at the base speed N_{base} . Reducing armature voltage provides a constant torque speed control where the speed can be reduced below the base value, while the motor has full torque capability. For obtaining speeds above N_{base} the field is gradually weakened. The motor torque therefore reduces as its speed increases which corresponds to constant kW (or HP) drive. This is shown in the Figure 2.29.

Some of the features of the Ward Leonard system are given below:

1. Absence of external resistance improves efficiency at all speeds and also when the generator emf becomes less than the back emf of the motor, the electrical power flows back from motor to generator, is converted to mechanical form and is returned to the mains via the driving AC motor.
2. Motor starts up smoothly therefore No starting device is required.
3. Speed reversal is smoothly carried out.
4. Fine speed control from zero to rated value in both the directions.

This method of speed control is used in

- a. High speed elevators
- b. Colliery winders

Advantages

1. Absence of external resistance improves efficiency at all speeds
2. Motor starts up smoothly. No starting device is required
3. Speed reversal is smoothly carried out

STARTERS FOR DC MOTORS

Necessity of starter:

The current drawn by the armature is given by $I_a = \frac{V - E_b}{R_a}$. At starting, $E_b = 0$ because $N = 0$.

Armature resistance will be very low. Therefore, the current drawn by the motor will be very high. In order to limit this high current, a starting resistance is connected in series with the armature. The starting resistance will be excluded from the circuit after the motor attains its rated speed. From there on back emf limits the current drawn by the motor. The arrangement is shown in the figure 2.30 which is a three point starter for shunt motor.

Three Point Starter:

It consists of resistances arranged in steps, R_1 to R_5 connected in series with the armature of the shunt motor. Field winding is connected across the supply through a protective device called 'NO – Volt Coil'. Another protection given to the motor in this starter is 'over load release coil'. The arrangement is shown in Figure 2.30

To start the motor the starter handle is moved from OFF position to Run position gradually against the tension of a hinged spring. An iron piece is attached to the starter handle which is kept hold by

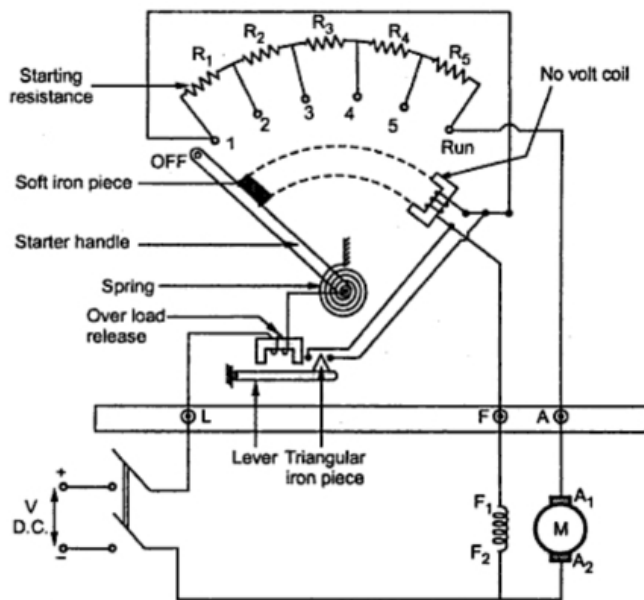


Figure 2.30

the No-volt coil at Run position. The function of No volt coil is to get deenergised and release the handle when there is failure or disconnection or a break in the field circuit so that on restoration of supply, armature of the motor will not be connected across the lines without starter resistance. If the motor is over loaded beyond a certain predetermined value, then the electromagnet of overload release will exert a force enough to attract the lever which short circuits the electromagnet of No volt coil. Short circuiting of No volt

coil results in deenergisation of it and hence the starter handle will be released and return to its off position due to the tension of the spring. In this type of starter, the shunt field current has to flow back through the starter resistance thus decreasing the shunt field current. This can be avoided by placing a brass arc on which the handle moves as shown in Figure 2.31.

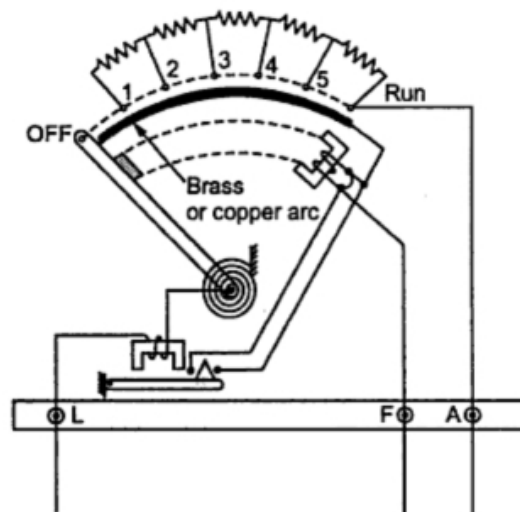


Figure 2.31

FOUR POINT STARTER: Figure 2.32 shows a four point starter.

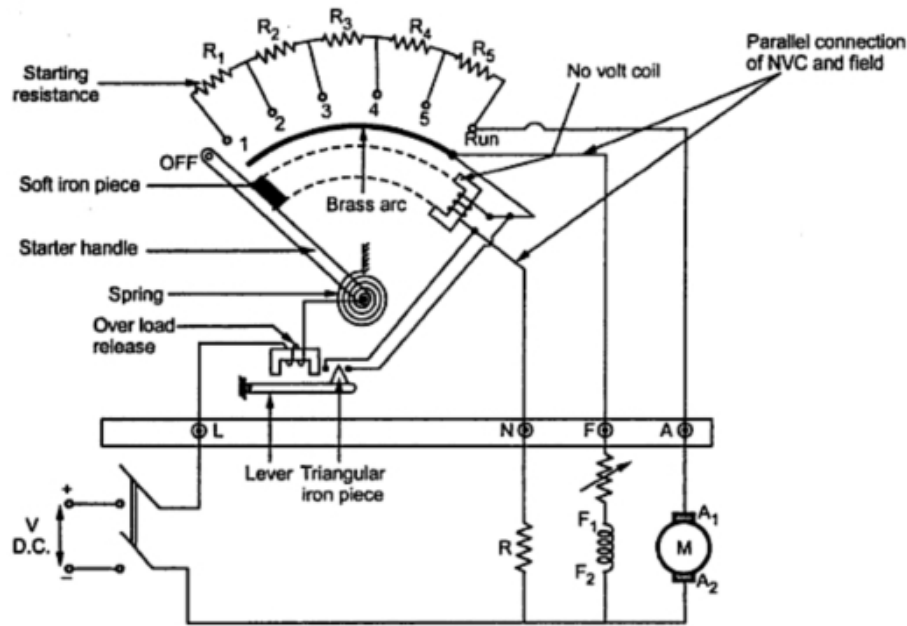


Figure 2.32

One important change is the No Volt Coil has been taken out of the shunt field and has been connected directly across the line through a Protecting resistance 'R'. When the arm touches stud one. The current divides into three paths, 1. Through the starter resistance and the armature, 2. Through shunt field and the field rheostat and 3. Through No-volt Coil and the protecting resistance 'R'. With this arrangement, any change of current in shunt field circuit does not affect the current passing through the NO-volt coil because, the two circuits are independent of each other. Thus the starter handle will not be released to its off position due to changes in the field current which may happen when the field resistance is varied.

Application of DC Motors:

DC Series Motor:

1. Starting torque is very high up to the five times the full load torque.
2. Speed regulation of the DC series motor can be varied widely.
3. Are used in hoists, cranes, battery powered vehicles, electric traction, etc.

DC Shunt Motor:

1. Has a medium starting torque.
2. Speed regulation is 5 to 15%.

3. It is essentially used for constant speed applications requiring medium starting torques such as centrifugal pumps, fans, blowers, conveyors, machine tools, printing presses etc.

DC Compound Motor:

1. Its characteristics depend on degree of compounding.
2. It has considerably high starting torque and drooping speed - load characteristics.
3. Used for varying loads such as shears, conveyors, crushes, punch presses, hoists, rolling mill, milling machines etc.

PROBLEMS:

1. A 250 V DC shunt motor has shunt field resistance of $150\ \Omega$ and armature resistance of $0.6\ \Omega$. The motor operates on No-load with a full field flux at its base speed of 1000 rpm with armature current of 5A. If the machine drives a load requiring a torque of 100Nm, calculate the armature current and speed of motor. If the motor is required to develop 10 kW at 1200 rpm, what is the required value of the external series resistance in the field circuit? Neglect saturation and armature reaction.

SOLUTION:

We know that, $E_a = K_a \Phi \omega_m = K \omega_m$

$$T = K_a \Phi I_a = K I_a; \text{ At No-load; } 250 - 5 \times 0.6 = K \times \frac{2\pi \times 1000}{60} \text{ Or } K = 2.36$$

When driving a load of 100 Nm,

$$I_a = \frac{T}{K} = \frac{100}{2.36} = 42.4\ A$$

$$\text{Now, } \omega_m = \frac{E_a}{K} = \frac{250 - 42.4 \times 0.6}{2.36} = 95.15 \text{ rad/s; } n = \frac{60}{2\pi} \omega_m = 909\ \text{rpm}$$

Given: Output = 10 kW at 1200 rpm

Assuming linear magnetization, it can be written as

$$E_a = K'_a I_f \omega_m; \quad T = K'_a I_f I_a;$$

From the data of No-load operation

$$I_f = \frac{250}{150} = 1.67\ A;$$

$$K'_a = \frac{K}{I_f} = \frac{2.36}{1.67} = 1.413;$$

$$\text{Now } (250 - 0.6 I'_a) I'_a = 10 \times 1000$$

Solving $I'_a = 44.8\ A$ (the higher value is rejected)

Substituting the values in the above equations,

$$250 - 0.6 \times 44.8 = 1.413 \times I_f' \times \frac{2\pi \times 1200}{60}$$

$$\text{Or } I_f' = 1.257 \text{ A}$$

$$\text{Therefore } R_f(\text{total}) = \frac{250}{1.257} = 199 \Omega$$

$$\text{Hence } R_f(\text{external}) = 199 - 150 = 49 \Omega.$$

2. A 4-pole series wound fan motor draws an armature current of 50A, when running at 2000 rpm on a 230 V DC supply with four coils connected in series. The four field coils are now connected in two parallel groups of two coils in series. Assuming the flux/pole to be proportional to the exciting current and load torque proportional to the square of speed, find the new speed and armature current. Neglect losses. Given: Armature resistance = 0.2 Ω , resistance of each field coil = 0.05 Ω .

SOLUTION:

$$E_a = K_E I_{se} n$$

$$T_L = K_L n^2 = T_M = K_M I_{se} I_a;$$

$$\text{Field coils in series; } I_{se} = I_a; R_{se} = 4 \times 0.05 = 0.2 \Omega, R_a = 0.2 \Omega$$

$$R_A = 0.2 + 0.2 = 0.4 \Omega; I_{a1} = 50 \text{ A}$$

$$E_{a1} = 230 - 0.4 \times 50 = 210 \text{ V}$$

$$210 = K_E \times 50 \times 2000$$

$$K_L \times (2000)^2 = K_M \times (50)^2$$

Field coils in two series group in parallel

$$I_{se}(\text{effective}) = I_a$$

$$R_{se} = \frac{2 \times 0.05}{2} = 0.05 \Omega$$

$$R_A = 0.2 + 0.05 = 0.25 \Omega$$

$$E_{a2} = 230 - 0.25 I_{a2} = K_E \times \left(\frac{I_{a2}}{2}\right) n_2$$

$$K_L n_2^2 = K_M \times \left(\frac{I_{a2}}{2}\right) \times I_{a2}$$

From the above equations,

$$\frac{n_2^2}{(2000)^2} = \frac{I_{a2}^2}{2 \times (50)^2}$$

$$\text{If } n_2 = 28.3 I_{a2}$$

From the equations, we have

$$\frac{230 - 0.25 I_{a2}}{210} = \frac{I_{a2} n_2}{2 \times 50 \times 2000}$$

Substituting for n_2

$$I_{a2}^2 + 8.4 I_{a2} - 7740 = 0$$

$$I_{a2} = 83.9 \text{ A; ignoring negative value}$$

$$n_2 = 28.3 \times 83.9 = 2374 \text{ rpm}$$

3. A shunt motor runs at 1200 rpm on a 420 V circuit and current taken is 30 A in addition to the field current. What resistance must be placed in series with armature in order that the speed may be reduced to 600 rpm, the current through the armature remaining same. $r_a = 3 \Omega$ given.

SOLUTION: $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$

$$N_1 = 1200 \text{ rpm; } N_2 = 600 \text{ rpm}$$

$$E_{b1} = 420 - 30 \times 3 = 330 \text{ V; } E_{b2} = 420 - 30 R_t$$

$$\frac{600}{1200} = \frac{420 - 30 R_t}{330}; R_t = 8.5 \Omega$$

$$R_{ex} = R_t - r_a = 8.5 - 3 = 5.5 \Omega$$

4. A 4 pole 230 V series motor runs at 1000 rpm when the load current is 12 A. the series field resistance is 0.8Ω and the armature resistance is 1Ω . The series field coils are now regrouped from all in series to two in series with two parallel paths. The line current is now 20 A if the corresponding weakening of field is 15 %. Calculate the speed of the motor.

SOLUTION: $\frac{N_2}{N_1} = \frac{E_{b2} \phi_1}{E_{b1} \phi_2}$

$$E_{b1} = 230 - 12 \times 1.8 = 208.4$$

$$E_{b2} = 230 - 20 \left(1 + \frac{0.4}{2} \right) = 206 \text{ V}$$

$$\frac{N_2}{1000} = \frac{206}{208.4} \times \frac{1}{0.85} = N_2 = 1163 \text{ rpm.}$$

5. A 220 V DC shunt motor runs at 500 rpm when the armature current is 50 A calculate the speed if the torque is doubled. Given $r_a = 0.2 \Omega$.

SOLUTION: $T_a \propto \Phi I_a$

$$T_{a2} = 2 T_{a1}$$

Therefore $\frac{T_{a2}}{T_{a1}} = 2$

$$\frac{I_{a2}}{I_{a1}} = \frac{T_{a2}}{T_{a1}}$$

$$\Rightarrow I_{a2} = 100 \text{ A}$$

Using $E_b = V - I_a r_a$

$$E_{b2} = 200 \text{ V}$$

$$E_{b1} = 210 \text{ V}$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$$

$$\Rightarrow N_2 = 476 \text{ rpm.}$$

6. A 500 V, 50 BHP (37.8 kW) 1000 rpm DC shunt motor has, on full load an η of 90%. The armature circuit resistance is 0.24Ω and there is a total voltage drop of 2 V at the brushes. The field current is 1.8 A. Determine

- (i) Full load line current
- (ii) Full load shaft torque
- (iii) Total resistance in motor starter to limit the starting current to 1.5 times the full load current.

SOLUTION: (i) motor input $= \frac{37.8 \times 1000}{0.9} = 41444 \text{ W}$

Therefore full load line current $= \frac{41444}{500} = 82.9 \text{ A}$

(ii) $T_{sh} = \frac{\text{output in watts}}{2\pi N} = 356 \text{ N-m}$

(iii) Starting line current $= 1.5 \times I_{FL} = 124.3 \text{ A}$

$$I_a \text{ at starting} = 124.3 - 1.8 = 122.5 \text{ A}$$

Let R = Starter Resistance

$$122.5 (R + 0.24) + 2 = 500 \text{ V}$$

$$R = 3.825 \Omega.$$

7. A 460 V series motor runs at 500 rpm taking a current of 40 A. Calculate the speed and % change in torque if the load is reduced, so that the motor is taking 30 A. Total resistance of the armature and field circuit is 0.8Ω . Assume flux is proportional to the field current.

SOLUTION: $\Phi \propto I_a$, $T_a \propto \Phi I_a \propto I_a^2$

$$\frac{T_2}{T_1} = \frac{9}{16} \quad \% \text{ change in torque} = \frac{T_2 - T_1}{T_1} \times 100 = 43.75\%$$

Using $E_b = V - I_a r_a$, $E_{b1} = 428.6 \text{ V}$, $E_{b2} = 436 \text{ V}$.

$$\frac{N_2}{N_1} = \frac{E_{b2} I_{a1}}{E_{b1} I_{a2}}, N_2 = 679 \text{ rpm.}$$

8. A 220 V DC series motor is running at a speed of 800 rpm and draws 100 A. Calculate at what speed the motor will run when developing half the torque. Armature + field resistance is 0.1Ω . Assume that magnetic circuit is unsaturated.

SOLUTION: $\frac{N_2}{N_1} = \frac{E_{b2} I_{a1}}{E_{b1} I_{a2}}$

$$\Phi \propto I_a, T_a \propto \Phi I_a \propto I_a^2$$

$$T_1 \propto I_{a1}^2$$

$$T_2 \propto I_{a2}^2$$

$$I_{a2} = 70.7 \text{ A}$$

Using $E_b = V - I_a r_a$

$$E_{b1} = 210 \text{ V}, E_{b2} = 212.9 \text{ V}$$

$$\frac{N_2}{N_1} = \frac{E_{b2} I_{a1}}{E_{b1} I_{a2}} \Rightarrow N_2 = 1147 \text{ rpm.}$$

9. An 8kW 230 V, 1200 rpm DC shunt motor has $r_a = 0.7 \Omega$. the field current is adjusted until, on NO load with a supply of 250 V, the motor runs at 1250 rpm and draws a armature current of 1.6 A. A load torque is then applied to the motor shaft, which causes the armature current to raise to 40 A and the speed falls to 1150 rpm. Determine the reduction in flux per pole due to the armature reaction.

SOLUTION: $N \propto \frac{E_b}{\Phi}$ or $N = K_N \left(\frac{V - I_a r_a}{\Phi} \right)$ or $\Phi = K_N \left(\frac{V - I_a r_a}{N} \right)$

$$\Phi_{(\text{no load})} = 0.2 K_N$$

$$\Phi_{(\text{load})} = 0.193 K_N$$

$$\text{Reduction in flux/pole due to armature reaction} = \frac{0.2 - 0.193}{0.2} \times 100 = 3.5\%$$

10. A 230 V, DC shunt motor runs at 800 rpm and takes armature current of 50 A. Find the resistance to be added to the field circuit to increase the speed from 800 rpm to 1000 rpm at an armature current of 80 A. Assume the flux to be proportional to the field current. Given $r_a = 0.15\Omega$ and $r_f = 250\Omega$.

SOLUTION: $I_{f1} = \frac{V}{r_f} = \frac{230}{250} = 0.92 \text{ A}, \quad I_a = 50 - 0.92 = 49.08 \text{ A}$

Using $E_b = V - I_a r_a$; $E_{b1} = 222.64 \text{ V}; \quad E_{b2} = 218 \text{ V}$

$\Phi \propto I_f, \quad I_{f2} = \frac{V}{r_f + R} \quad \text{taking } r_f + R = R_t, \text{ and using}$

$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \cdot \frac{I_{f1}}{I_{f2}}, \quad R_t = 250 + R = 319 \Omega; \quad R = 69\Omega.$

SPECIAL MACHINESPermanent magnet D.C motors:

These are same as that of ordinary D.C shunt motors, but the difference is they have permanent magnets instead of stationary field winding for producing the required magnetic flux.

Construction:

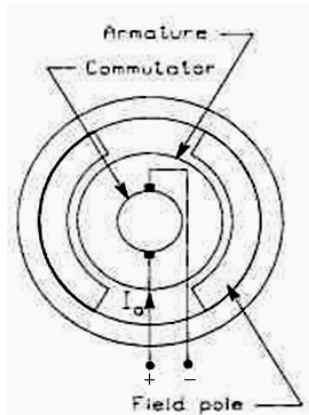


Figure 2.33

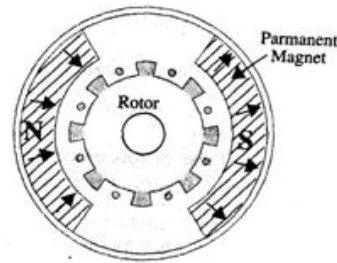


Figure 2.34

The Figure 2.33 and 2.34 Shows the construction of a permanent magnet D.C motor. The magnetic frame or yoke supports the permanent magnet & also produces the return path for magnetic flux.

The armature consists of slot for windings, commutator segments & brushes are same as those in conventional D.C motors. The magnets are made of materials like ferrite, alnico, iron-cobalt, etc. These materials have high residual flux-density, & high energy output. These materials also have good mechanical properties.

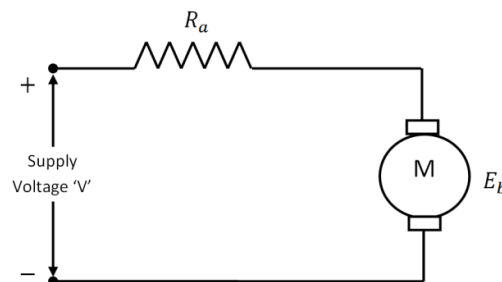
Working & performance characteristics:

Figure 2.35

The equivalent circuit of a permanent magnet D.C motor is shown in Figure 2.35

R_a = Arm resistance; field windings are absent of permanent magnets.

$$E_b = \frac{\Phi Z n P}{60 A} = \frac{\Phi Z n}{A} ; \quad n \text{ is in rps} = \frac{N}{60}$$

$$E_b = \frac{Z P \Phi n}{A} \times \frac{2\pi}{2\pi} = \frac{Z P}{A} \Phi \frac{\omega}{2\pi}$$

$$\text{Therefore, } E_b = \frac{Z P}{A 2\pi} \omega \Phi \text{ or } E_b = K \times \Phi \times \omega \dots\dots\dots 36$$

$$\& T = \frac{1}{2\pi} \Phi Z I_a \frac{P}{A} \text{ or } T = K \Phi I_a \text{ -- in conventional motors.}$$

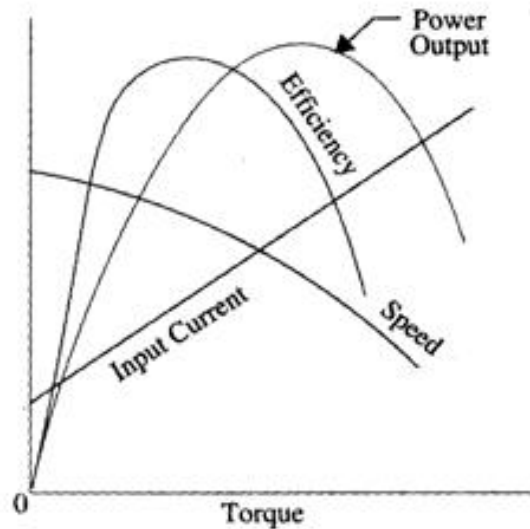
In case of permanent magnet D.C motors,

The resistant flux Φ is constant.

$$E_b = K \Phi \omega \text{ or } E_b = K_p \omega \text{ where } K_p = K \cdot \Phi$$

$$K_p = \frac{E_b}{\omega} \& \text{Torque } T = \frac{E_b I_a}{\omega} = K_p I_a \text{ (because } \frac{E_b}{\omega} = K_p \text{)}$$

$$V = E_b + I_a R_a = K_p \omega + I_a R_a. \quad \text{Therefore } \omega = \frac{V - I_a R_a}{K_p} \dots\dots\dots 37$$

Performance characteristics:**Figure 2.36**

Torque/s speed is almost linear. Therefore they are used in servo motors.

Efficiency is better than conventional motors. Therefore field losses are absent.

Because of constant flux, speed control is not possible with flux control method. Therefore only Armature control can be employed & hence the speeds obtained will be below the normal speeds.

Advantages:

1. No external Excitation is required to produce the flux. Therefore less losses & efficiency is high.
2. As field windings are absent, their size is smaller, cost decreases.
3. Motors designed upto 12V or less produce less TV & radio interference.
4. Less noise.

Disadvantages:

1. If the Armature reaction is more because of high armature current, permanent magnet may get demagnetized. Temperature may also affect the same.
2. The flux density produced in the air gap by the permanent magnet is limited.
3. Speeds above normal speeds are not possible.

Applications:

1. These motors normally run on 6V, 12V, or 24V D.C supply.
2. They are extensively used in automobiles.
3. They are used in blowers, heaters & air conditions.
4. They are used for disc drives in personal computers.
5. They are used in fans, radio antennas, marine engine starters, wheel chairs & cordless power tools, vacuum chamber, mixer etc.
6. They are available upto a rating of 150KW.

PROBLEM:

(1) A permanent magnet D.C motor has resistance of 1Ω , the speed of the motor is 2000rpm when fed from 50V D.C source while taking 1.2A determine

- (i) No load rotational losses.
- (ii) The motor output when running at 1800rpm when source voltage is 48V.
- (iii) Stalling torque when fed from 20V source.

SOLUTION:

$$E_b = V - I_a r_a = 50 - 1.2 \times 1 = 48.8V$$

$$(i) \quad \text{At no load, no load losses} = E_b I_a = (48.8) \times (1.2) = 58.56 \text{ W.}$$

$$(ii) \quad E_b = K_p \omega, K_p = E_b / \omega = 48.8 / 2\pi(2000/60) \\ = 0.2330 \text{ V-S/rad.}$$

For a speed of 1800rpm,

$$\omega = 2\pi N/60 = 2\pi (1800)/60 = 188.49 \text{ rad/s.}$$

$$\text{Therefore } E_b = K_p \omega = (0.2330) (188.49) = 43.91V$$

$$I_a = (V - E_b) / r_a = (50 - 43.91) / 1 = 6.09A$$

$$\text{Power developed} = E_b I_a = (43.91) (6.09) = 267.59 \text{ W}$$

$$\text{Motor output} = E_b I_a - \text{Rotational losses} = 267.59 - 58.56 = 209.03 \text{ W}$$

$$(iii) \quad E_b = 0 \text{ when motor stalls, } V = I_a r_a$$

$$\text{Therefore } I_a = V / r_a = 20 / 1 = 20A$$

$$\text{Stall torque} = K_p I_a = (0.2330) (20) = 4.66 \text{ N-m}$$

Brushless DC Motor:

Construction of brushless DC motor:

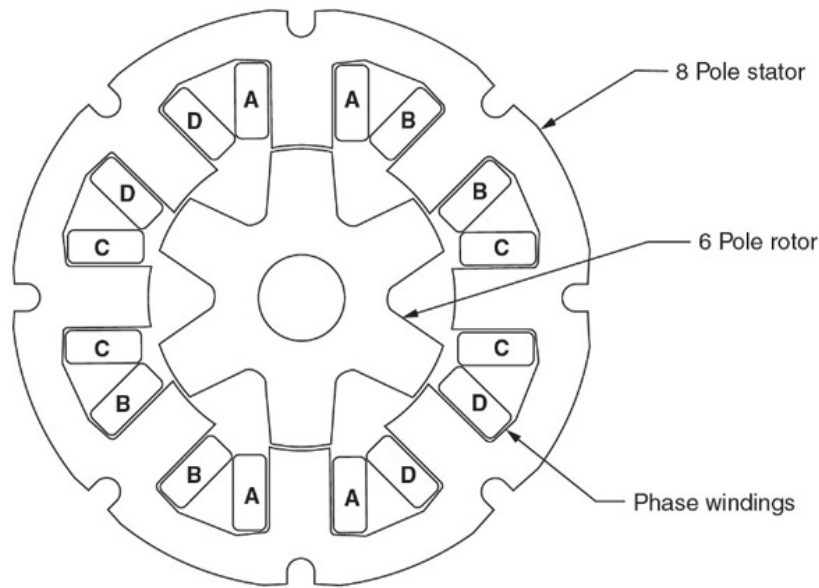


Figure 2.37

The construction of brushless DC motor is shown in Figure 2.37. It consists of a stator and a permanent magnet rotor. The stator houses a polyphase winding in its slots. The rotor consists of permanent magnets. The stator windings are fed by an inverter. Rotor position sensors generate pulses for controlling the transistors in the inverter.

There are 2 types Brushless DC Motor

1. Unipolar or half wave Brushless DC Motor

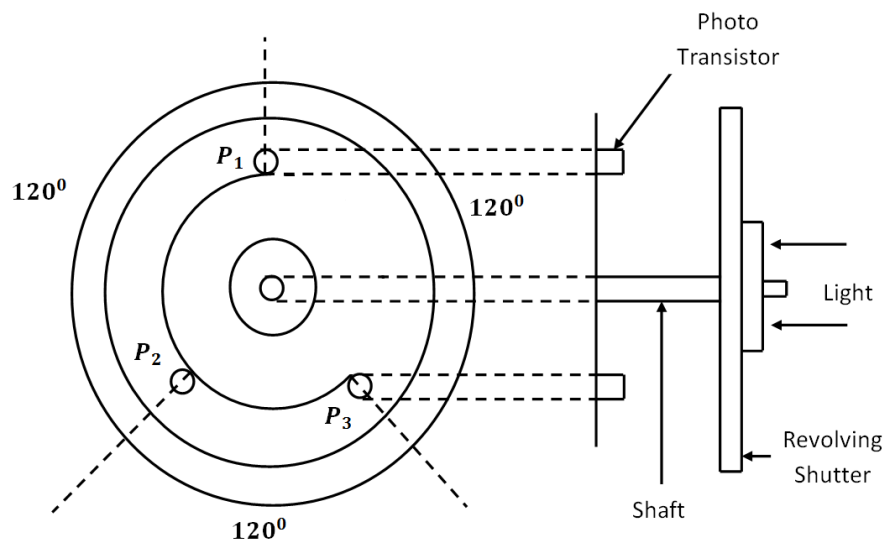


Figure 2.38

In this type, rotor consists of optical sensors. The optical sensor has a light source, three phototransistors P_1 , P_2 and P_3 mounted on the end plate of the motor separated by 120° from each other and a revolving shutter coupled to the shaft of the motor. The optical sensor is shown in Figure 2.38. The stator, two pole rotor and driving circuit used to excite the stator winding are shown in Figure 2.39.

When the light falls on P_1 it actuates the transistor Q_1 and a pulse (Pl_1) appears across Ph_1 (phase 1) and creates a magnetic pole say North. The south pole of the rotor is attracted by this Ph_1 and the rotor aligns with it. As the revolving shutter is coupled to shaft of the rotor, the light falls on P_2 actuating transistor Q_2 resulting in a pulse (Pl_2) appearing across Ph_2 . This pulse creates a pole and the rotor is attracted by this Ph_2 . In this way, the rotor rotates in Uni-polar brushless DC motor.

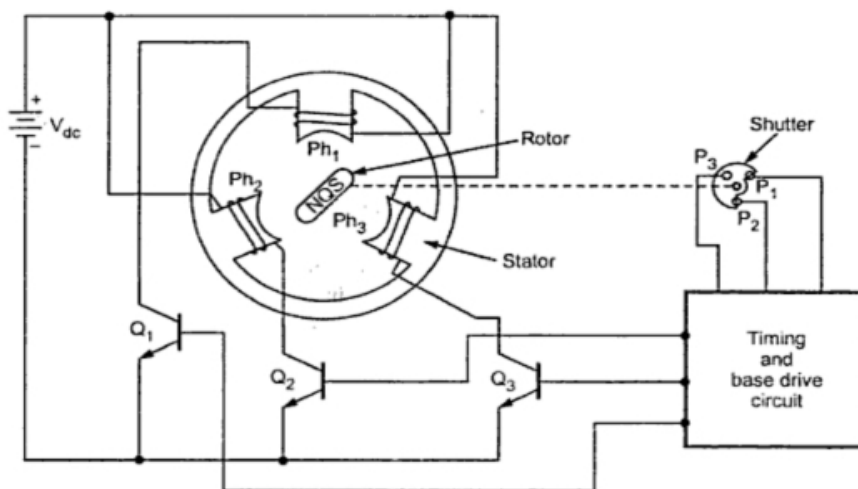


Figure 2.39

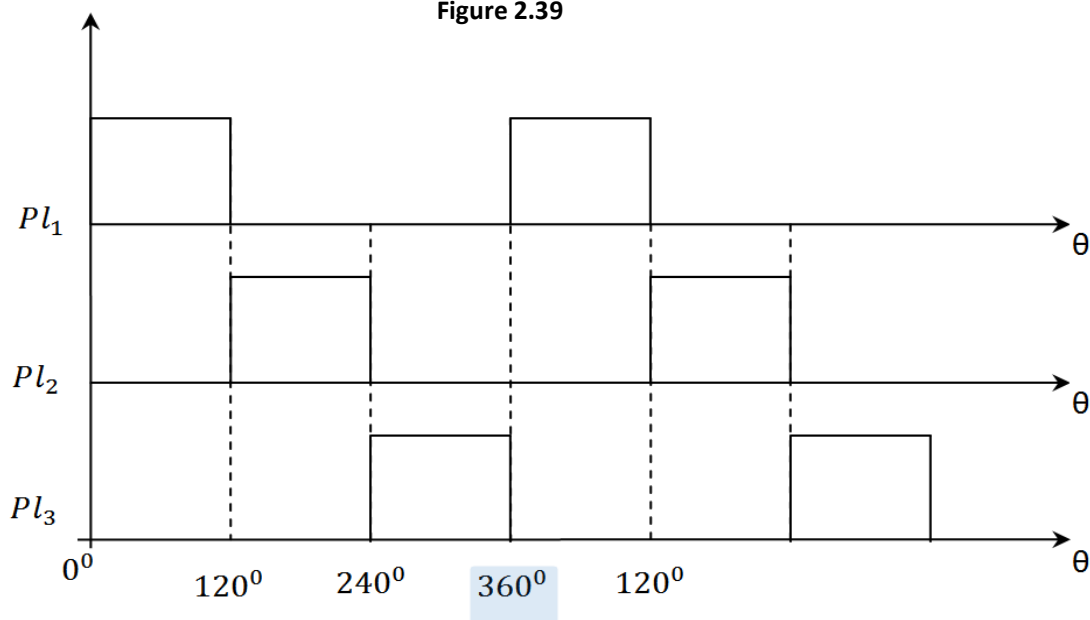


Figure 2.40

2. Bipolar Brushless DC Motor

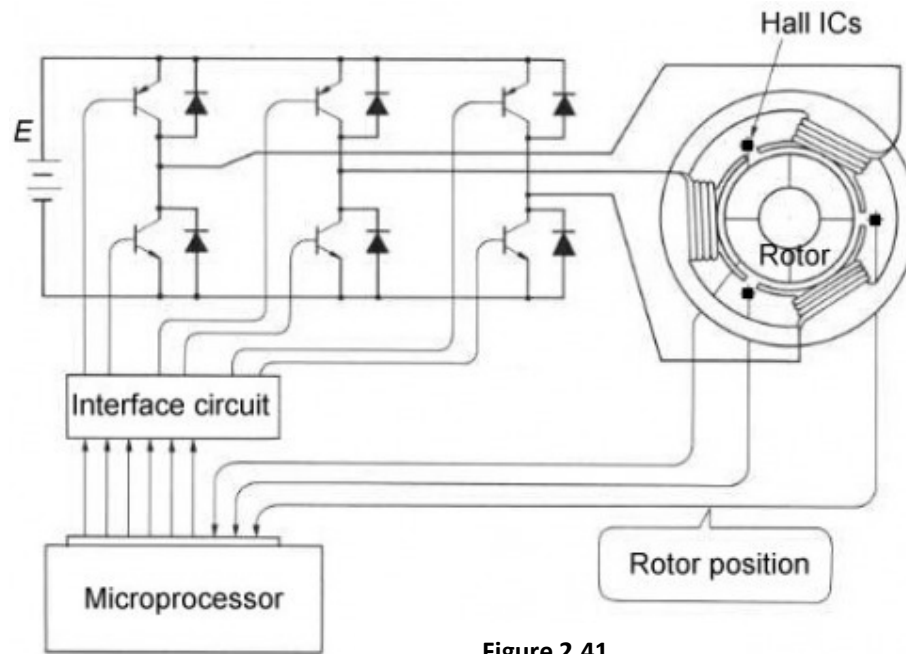


Figure 2.41

Figure 2.41 shows the bipolar brushless DC motor. In the motors of higher ratings the inductive energy due to the windings will be high. This motor uses three phase inverter with feedback diodes to return the inductive energy to the supply. The rotor uses Hall effect sensors to sense the rotor position. The hall sensors are 120° electrical from each other. Switching of each transistor occurs in response to rotor position sensor. Transistors are triggered in sequence. A pair of transistor conducts for 120° . Torque reversal is achieved by the shifting the base drive signal by 120° . This motor is suitable for high performance servo drives and also for ratings higher than 100 Watts.

Features of brushless DC motor:

- i) Due to absence of brushless & commutator they require practically no maintenance.
- ii) The operation is highly reliable and they have long life.
- iii) They have low inertia & friction.
- iv) Low radio frequency interference & hence the operation is noiseless.
- v) Speeds upto 30000rpm & more is possible.
- vi) Cooling is much better.
- vii) Due to feed back diodes efficiency is high (75-80%).

Applications of brushless DC motor:

- i) Unipolar **brushless** DC motor upto 100W, table driver, record players, spindle drives in H D drives, video recorders control systems.
- ii) Also in aerospace, gyroscope & biomedical instruments.